Nishant Singh

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Education

Indian Institute of Information Technology Bhopal — Bhopal, MP
B.Tech in Information Technology (GPA: 8.73) — *Dec 2021 – Jun 2025*
- **Coursework:** Data Structures and Algorithms, Database Management System, Operating System, Computer Network

Experience

Software Engineer — Jul 2025 – Present
Ultimate Kronos Group — *Pune, MH*
- Designed and built a **Playwright** test framework and CI pipeline from scratch with reusable fixtures, page objects, and automated reporting, cutting regression runtime by **86.7%** from **15 hours to 2 hours** and reducing flakiness by **54%**
- Developed compensation **hierarchy** views for plans with up to **20,000 employees** using **Angular, Spring Boot**, and **PostgreSQL**, implementing multi-currency aggregation, reporting structure validation, and hierarchy synchronization
- Remediated **Kafka** failure handling in salary plan distribution, preventing **producer** errors from stalling workflows, and implemented audit event publishing for tracking and auditing compensation changes across salary planning workflows
- Optimized **GKE** infrastructure and retired obsolete Helm deployment paths post-migration to **Cloud Run**, resolving CI dependencies and eliminating up to **$250/month** in development environment costs across **12 namespaces** in GKE

Mentor — Mar 2025 – Nov 2025
Google Summer of Code — *Remote*
- Mentored contributors to design and implement a scalable observability layer for KubeArmor, integrating **6 Prometheus** metrics to expose real-time insights into policy enforcement, alerting, and enhanced security activity across workloads
- Introduced and exposed **4+** key production-grade metrics endpoints in **Go**, adhering to Prometheus exposition standards and enabling integration with monitoring tools like **Grafana** and **Alert Manager**, leading to improved **observability**

Software Developer Intern — Jan 2025 – May 2025
Samsung Research Institute - Delhi — *Noida, UP*
- Implemented **runtime** API validation using the validator package in **Go**, adding field-level constraints like range values to Swagger specifications for **11 Iris-based** backend APIs, reducing **80%** of validation errors and improving API reliability
- Revamped UMD movie matching by manually mapping unmatched metadata entries between **Samsung**'s UMD and IMDb database, and developing a custom CLI tool that reduced the task time from **4–5 hours** to just **30 minutes** daily

DevOps Engineer Intern — Oct 2023 – Oct 2024
GoComet Pvt Ltd — *Bangalore, KA*
- Optimized **Fluentd** log ingestion and **New Relic** telemetry by enriching application logs and eliminating unnecessary metrics collection, reducing observability costs by up to **45%** while maintaining monthly spend within the **$2,000** budget
- Spearheaded daily deployment process of production code using an enhanced rolling deployment strategy on Kubernetes, orchestrating updates across **5 systems** and leveraged automated tools to minimize downtime and enhanced reliability
- Established a robust monitoring infrastructure by configuring alerts for **PostgreSQL 13**, specifically targeting long idling transactions, locks, and max concurrent connections, providing visibility into potential issues and enabled swift resolution

Projects

Blaze | *Go, Next.js, PostgreSQL, WebSocket, WebRTC* — github.com/tesla59/blaze
- Coded an Omegle-like real-time video chat platform using **Gorilla WebSockets** in **Go**, achieving **10,000+** concurrent WebSocket connections on a single core, following SOLID principles and a modular design for concurrent session handling
- Persisted **100,000+** user sessions, metadata, and abuse reports in **PostgreSQL**, building a **Next.js** frontend integrated with **WebRTC** to support **5,000+** simultaneous peer-to-peer video/audio streams through a custom Go signaling server

GoLoadBalancer | *Go, Docker, Load Balancer, Concurrency* — github.com/tesla59/GoLoadBalancer
- Developed a concurrent **Go** load balancer with round-robin routing, inspired by **Kubernetes** v1.24 operator architecture
- Orchestrated **3+** load balancers and **5+** workers using **Goroutines**, **Docker**, and **SQLite 3** for worker health metrics

Technical Skills

Languages: Go, Java, C, C++, TypeScript, JavaScript, Python
Frameworks & Testing: Spring Boot, Next.js, Flutter, Iris, Gorilla WebSocket, GoFiber, Playwright
Cloud & DevOps: AWS, Google Cloud, Azure, Docker, Kubernetes, Terraform, Ansible, Helm, GitHub Actions
Tools & Observability: Linux, Git, SQL, Kafka, Redis, gRPC, NGINX, Prometheus, Grafana, New Relic, Cloud IAM

Certifications

Google Cloud Certified Professional Cloud Architect | *Google Cloud* — Sep 2026